

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

33,641
held-out plate appearances

3.05%
held-out HR rate

0.6002
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-26, scored on the 33,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5861	1.53×	0.02951	0.13549	0.00512
KNN comps Prior is the pooled rate of the nearest comparables.	0.6002	1.51×	0.02947	0.13461	0.00446
SVR Support-vector regression on the same feature matrix.	0.5941	1.52×	0.02951	0.13545	0.00581
Random forest Forest regression; a learned similarity metric.	0.5942	1.49×	0.02950	0.13498	0.00500
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6002	1.51×	0.02947	0.13461	0.00446
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6002	1.51×	0.02947	0.13461	0.00446
Linear regression Ordinary least squares on the same features.	0.5946	1.53×	0.02949	0.13505	0.00431
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5951	1.62×	0.02951	0.13510	0.00611
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5946	1.57×	0.02950	0.13499	0.00552
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5948	1.52×	0.02951	0.13508	0.00595

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,663 PA, actual HR rate 2.46%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5600	1.11×	0.02398	0.11553	0.00855
KNN comps Prior is the pooled rate of the nearest comparables.	0.5714	1.33×	0.02403	0.11612	0.01453
SVR Support-vector regression on the same feature matrix.	0.5714	1.33×	0.02423	0.12067	0.01806
Random forest Forest regression; a learned similarity metric.	0.5818	1.45×	0.02415	0.11694	0.01637
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5714	1.33×	0.02403	0.11612	0.01453
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5714	1.33×	0.02403	0.11612	0.01453
Linear regression Ordinary least squares on the same features.	0.5735	1.11×	0.02414	0.11857	0.01553
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5763	1.45×	0.02415	0.11775	0.01611
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5756	1.33×	0.02410	0.11717	0.01479
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5808	1.33×	0.02421	0.11788	0.01802

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Ben Rice 19.6%, Luis García Jr. 16.8%, Munetaka Murakami 16.4%, Junior Caminero 16.4%, Spencer Jones 15.8%, George Lombard Jr. 14.7%
Flat league	6/6	Ben Rice 18.2%, Luis García Jr. 16.1%, Junior Caminero 16.0%, George Lombard Jr. 15.1%, Munetaka Murakami 14.9%, Spencer Jones 14.6%
KNN comps	6/6	Ben Rice 19.6%, Luis García Jr. 16.9%, Munetaka Murakami 16.6%, Junior Caminero 16.5%, Spencer Jones 15.8%, George Lombard Jr. 14.5%
SVR	6/6	Ben Rice 20.1%, Junior Caminero 17.7%, Luis García Jr. 17.3%, Munetaka Murakami 17.0%, George Lombard Jr. 14.9%, Spencer Jones 14.4%
Random forest	6/6	Ben Rice 19.8%, Luis García Jr. 17.2%, Junior Caminero 16.9%, Munetaka Murakami 16.6%, George Lombard Jr. 14.8%, Spencer Jones 14.8%
Core (KNN+SVR)	6/6	Ben Rice 19.6%, Luis García Jr. 16.9%, Munetaka Murakami 16.6%, Junior Caminero 16.5%, Spencer Jones 15.8%, George Lombard Jr. 14.5%
Forest blend	6/6	Ben Rice 19.6%, Luis García Jr. 16.9%, Munetaka Murakami 16.6%, Junior Caminero 16.5%, Spencer Jones 15.8%, George Lombard Jr. 14.5%
Linear regression	6/6	Ben Rice 19.5%, Luis García Jr. 17.0%, Junior Caminero 16.7%, Munetaka Murakami 16.3%, Spencer Jones 15.0%, George Lombard Jr. 14.9%
Logistic regression	6/6	Ben Rice 19.7%, Luis García Jr. 17.4%, Junior Caminero 17.0%, Munetaka Murakami 16.5%, Spencer Jones 15.0%, George Lombard Jr. 14.2%
Neural net (torch)	6/6	Ben Rice 19.7%, Luis García Jr. 17.3%, Junior Caminero 17.0%, Munetaka Murakami 16.6%, Spencer Jones 15.3%, George Lombard Jr. 14.5%
XGBoost	6/6	Ben Rice 19.7%, Junior Caminero 17.4%, Luis García Jr. 17.4%, Munetaka Murakami 17.0%, Spencer Jones 15.3%, George Lombard Jr. 14.9%

Hitter by hitter

The top 40 bats by the served probability.

ML Model Performance Metrics by Player											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T)	
19Ryan McMahon	YY	9.7%	9.0%	9.5%	9.3%	9.5%	9.5%	9.5%	9.3%		
20Chase DeLaeter	LE	8.9%	9.0%	8.8%	8.7%	8.7%	8.8%	8.7%	8.7%		
21Nathaniel Lowe	LE	8.8%	8.5%	8.9%	8.9%	9.1%	8.9%	8.9%	8.7%		
22Tristan Peters	CWS	8.7%	9.2%	8.9%	9.0%	8.9%	8.9%	8.9%	9.0%		
23Angel Martínez	CLE	8.4%	8.7%	8.3%	8.6%	8.5%	8.3%	8.3%	8.4%		
24Patrick Bailey	CLE	8.1%	7.6%	8.1%	7.8%	7.6%	8.1%	8.1%	7.9%		
25Sam Antonacci	CWS	8.0%	9.3%	7.9%	7.0%	8.1%	7.9%	7.9%	7.8%		
26Jonny DeLuca	TB	7.7%	8.6%	8.0%	8.0%	7.5%	8.0%	8.0%	7.8%		
27Travis Bazzana	CLE	7.2%	7.4%	7.1%	7.1%	7.3%	7.1%	7.1%	7.1%		
28Braden Montgomery	NY	7.1%	8.1%	6.8%	6.5%	6.7%	6.8%	6.8%	6.7%		
29José Ramírez	CLE	7.0%	7.3%	7.0%	6.8%	7.1%	7.0%	7.0%	7.3%		
30Liam Hicks	TB	7.0%	8.0%	6.9%	7.8%	7.4%	6.9%	6.9%	7.4%		
31Chase Meidroth	CWS	6.9%	7.5%	6.7%	7.2%	7.0%	6.7%	6.7%	6.9%		
32Richie Palacios	TB	6.1%	6.7%	6.0%	5.9%	5.7%	6.0%	6.0%	5.9%		
33Bryan Rocchio	CLE	5.3%	5.7%	5.3%	5.4%	5.3%	5.3%	5.3%	5.1%		
34Taylor Walls	TB	4.0%	5.5%	4.2%	3.5%	3.4%	4.2%	4.2%	4.1%		
35Steven Kwan	CLE	3.6%	6.2%	3.7%	2.1%	2.8%	3.7%	3.7%	2.0%		
36Chandler Simpson	TB	2.9%	4.9%	2.9%	1.3%	1.3%	2.9%	2.9%	1.2%		

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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